BLOWING UP A FINITE TYPE IDEAL NEAR A CURVE POINT WHERE THE RESTRICTION IS AMPLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 23391-23400 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 23402-23438, the complete author proof (35 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Terminal hub of the chapter's blow-up series; the only \cite in the collected spans, satisfied by a card-local bibliography.tex (Gross-thesis entry from the snapshot my.bib).

Editorial note. The cited proposition's number appears verbatim in the retained statement, so the reference list is load-bearing for a reader; entry transcribed exactly from my.bib lines 2654-2660.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Gross, Vector bundles as generators on schemes and stacks (Proposition 1.3.1, an ampleness criterion on blow-ups checked formally along the centre), the chapter's preceding blow-up/slice lemmas used inside the proof, and standard ampleness machinery on the generic fibre.

Source reference. Original lemma 0809: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23438–23438. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02OS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23438–23438. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02NS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23438–23438. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 080A: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23438–23438. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0GTS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23361–23367. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZX: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23438–23438. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01PH: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23438–23438. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. [1, Proposition 1.3.1] *Let X be a quasi-compact and quasi-separated scheme with finitely many irreducible components. There exists a quasi-compact dense open $U \subset X$ and a U-admissible blowing up $X' \to X$ such that the scheme X' has an ample family of invertible modules.*

Proof. Let $\eta_1, \ldots, \eta_n \in X$ be the generic points of the irreducible components of X. By Properties, Lemma 01ZX and the fact that X is quasi-compact we can find a finite affine open covering $X = U_1 \cup \ldots \cup U_m$ such that each U_i contains $\eta_1, \ldots, \eta_n$. In particular the quasi-compact open subset $U = U_1 \cap \ldots \cap U_m$ is dense in X. Let $\mathcal{I}_i \subset \mathcal{O}_X$ be a finite type quasi-coherent ideal sheaf such that $U_i = X \setminus Z_i$ where $Z_i = V(\mathcal{I}_i)$, see Properties, Lemma 01PH. Let

$$f : X' \longrightarrow X$$

be the blowing up of X in the ideal sheaf $\mathcal{I} = \mathcal{I}_1 \cdots \mathcal{I}_m$. Note that f is a U-admissible blowing up as $V(\mathcal{I})$ is (set theoretically) the union of the Z_i which are disjoint from U. Also, f is a projective morphism and $\mathcal{O}_{X'}(1)$ is f-relatively ample, see Divisors, Lemma 02NS. By Divisors, Lemma 080A for each i the morphism f' factors as $X' \to X'_i \to X$ where $X'_i \to X$ is the blowing up in $\mathcal{I}_i$ and $X' \to X'_i$ is another blowing up (namely in the pullback of the products of the ideals $\mathcal{I}_j$ omitting $\mathcal{I}_i$). It follows from this that $D_i = f^{-1}(Z_i) \subset X'$ is an effective Cartier divisor, see Divisors, Lemmas 0809 and 02OS. We have $X' \setminus D_i = f^{-1}(U_i)$. As $\mathcal{O}_{X'}(1)$ is f-ample, the restriction of $\mathcal{O}_{X'}(1)$ to $X' \setminus D_i$ is ample. It follows from Lemma 0GTS that X' has an ample family of invertible modules. □

References

[1] Gross, Philipp. *Vector bundles as generators on schemes and stacks.* Düsseldorf, Univ., Diss., 2010.